

Report184

Fixed positive permanent binary matrices

OEIS A089479

Asymptotics, canonical block laws, and reproducible certificates

3 October 2026

Abstract

For each fixed positive integer k , let $C_{n,k}$ count all labelled binary $n \times n$ matrices with ordinary integer permanent k . We prove

$$C_{n,k} = \frac{(n!)^2}{k} 2^{\binom{n}{2}} \{ \rho^{-n} P_k(n) + O_k(2^{-n}) \},$$

where $\rho = 1.4880785455997102946562460\dots$ is the classical directed-acyclic-graph root, and P_k is an effectively computable polynomial of exact degree $\Omega(k)$, the number of prime factors of k with multiplicity. Its positive leading coefficient is an explicit product of prime amplitudes. The proof combines marked perfect matchings, finite suppressed strongly connected kernels, the classical Robinson component transform, and a self-contained complex root isolation. We compute the complete polynomials for $k = 2, 3, 4$ with rationally certified coefficient intervals, prove explicit all- n remainder constants, give every fixed Stirling and finite spectral order, and derive smooth and integer threshold inverse bounds. Canonical allowed-edge blocks have a limiting independent prime-block size law, with expansions to every fixed order in exponentially weighted total variation. Exact counting code and rational certificates are supplied separately from numerical diagnostics.

Scope and attribution. The model is the full permanent-distribution triangle A089479, with no row-degree or column-degree restriction. The $k = 1$ case is the known permanent-one/DAG correspondence, not a new counting theorem. The SCC transform is classical Robinson machinery, read in its primary 2019 restatement. The target-specific results are derived below; the bounded source review in Section 14 is not a worldwide priority determination. All limits keep k , every requested expansion order, and every probability weight fixed.

[write] **Status and trust boundary, added 6 October 2026, batch 108.** This is bundle Report 184 of an external AI-assisted research session, filed in ProveIt as the single-source research report `a089479-fixed-permanent-matrices` (provenance, the sources as the write read them, what was checked, relation to the repository, the collected non-claims and the reading conventions are in Section 1.2). It is unrefereed and not formalized. Its author line is a subtitle (“Asymptotics, canonical block laws, and reproducible certificates”) and its PDF author field reads “Research report”: it names no person or tool. *What rests on what.* Theorems 1.1 and 1.3 (through Theorem 10.1) and Theorems 9.1 and 12.1 have conventional proofs: the marked-matching identity (7), the circulation bound of Lemma 3.1, the finite kernel formula of Proposition 3.2, Robinson’s component transform (19) (classical, credited, and re-derived in Section 4 in its marked form), the root isolation of Lemma 5.1, the positivity identity of Lemma 6.1, and Cauchy estimates. No proof of these uses a computation. The rational functions (15)–(16) come from enumerating finitely many kernels, and the explicit constants of Theorem 1.2 and the effective inverse brackets (50) also use the rational interval

enclosures of Section 7, which are the output of the shipped exact program `code/certify_constants.py`; the remaining arithmetic of Section 8 is exact and was redone by hand at intake. The tables of $\pi_2(m)$ and of finite-size discrepancies are decimal diagnostics. The case $k = 1$ is the classical permanent-one/acyclic digraph correspondence, and its asymptotic is recorded in OEIS A003024 (Remark 1.4). The source promises a “second independent rational implementation” that is not in the package (note at the end of Section 7.3). The write adds Remark 1.4 (the four OEIS entries the source cites, quoted; the case $k = 1$ as an instance of the asymptotic stated in A003024), Remark 12.2 (the inverses of Sections 11–12 against the transseries volume), dated notes, and one question in Section 15. No statement, proof or number of the source was changed.

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1 Main results and normalization

For a binary matrix $A = (a_{ij})_{1 \leq i, j \leq n}$, define

$$\text{per } A = \sum_{\sigma \in \text{Sym}(n)} \prod_{i=1}^n a_{i, \sigma(i)}, \quad C_{n,k} = \#\{A \in \{0,1\}^{n \times n} : \text{per } A = k\}.$$

Rows and columns are labelled. We set $\text{per}(\emptyset) = 1$, so $C_{0,1} = 1$ and $C_{0,k} = 0$ for $k \neq 1$. In this report $k \geq 1$ is fixed. Write

$$k = \prod_p p^{e_p}, \quad d = \Omega(k) = \sum_p e_p, \quad E(z) = \sum_{m \geq 0} \frac{(-z)^m}{m! 2^{\binom{m}{2}}}.$$

The product is empty and $d = 0$ when $k = 1$. Section 5 proves that E has exactly one zero in $|z| \leq 2$, a simple real zero $\rho \in (1.48, 1.50)$. Put

$$a = -\rho E'(\rho) = \rho E(\rho/2) > 0. \quad (1)$$

The derivative identity follows by termwise differentiation of the entire series.

Theorem 1.1 (Fixed permanent asymptotic). *For every fixed integer $k \geq 1$ there is an effectively computable real polynomial P_k , of exact degree $d = \Omega(k)$, such that*

$$C_{n,k} = \frac{(n!)^2}{k} 2^{\binom{n}{2}} \{\rho^{-n} P_k(n) + O_k(2^{-n})\}. \quad (2)$$

Let $s_{m,p}$ count strongly connected loopless simple digraphs on m labelled vertices with $\text{per}(I + A) = p$. Antiparallel arcs are permitted. For each prime p , define

$$h_p = \sum_{m \geq 2} \frac{s_{m,p} \rho^m}{m! 2^{\binom{m}{2}}} E(\rho/2^m) > 0. \quad (3)$$

Then the leading monomial coefficient of P_k is

$$L_k = \frac{\prod_p h_p^{e_p}}{a^{d+1} \prod_p e_p!}. \quad (4)$$

In particular $P_1 = 1/a$. The O constant may depend on k .

The squared factorial in (2) is essential. One factor comes from labelled digraphs, the other from undoing a marked matching's column permutation. Neither the ordinary generating series $\sum C_{n,k} z^n$ nor its single-factorial exponential version is the meromorphic series used in the proof.

The conclusion identifies the complete dominant algebraic correction when factorials are retained: it is the finite polynomial P_k , not merely its leading monomial. For example, a prime k gives a linear polynomial, while $k = 4$ gives a quadratic. The degree is $\Omega(k)$, not $\lfloor \log_2 k \rfloor$.

Theorem 1.2 (Effective small permanent bounds). *Set $M_2 = 1000$, $M_3 = 3900$, and $M_4 = 170000$. For $k = 2, 3, 4$ and every integer $n \geq 0$,*

$$\left| \frac{k C_{n,k}}{(n!)^2 2^{\binom{n}{2}}} - \rho^{-n} P_k(n) \right| \leq M_k 2^{-n}. \quad (5)$$

The coefficients of P_k are certified in Section 7. Explicit increasing upper and lower smooth envelopes, and global threshold brackets derived from them, are valid on $x \geq 50$; see Section 11.3.

Theorem 1.3 (Prime block law). *In a uniformly chosen matrix counted by $C_{n,k}$, consider the connected components of the bipartite graph consisting only of entries that belong to at least one perfect matching. With probability $1 - O_k(n^{-1})$, exactly d components are nontrivial and their permanents are the prime factors of k , counted with multiplicity. For each prime p , the e_p limiting block sizes are independent with law*

$$\pi_p(m) = \frac{s_{m,p} \rho^m E(\rho/2^m)}{h_p m! 2^{\binom{m}{2}}}, \quad m \geq 2. \quad (6)$$

Their unordered multiset converges in total variation, and admits every fixed-order expansion in the exponentially weighted norm specified in Section 10. All these size laws have every positive exponential moment finite.

Thus the number of vertices in nontrivial allowed-edge blocks stays tight as $n \rightarrow \infty$. This does not say that the original matrix has only a bounded number of nonzero entries: entries outside the allowed-edge graph can be numerous.

1.1 Small exact values

The following counts use the OEIS convention at $n = 0$ and agree with the exact transform and the visible A089479 data [1]. They are useful normalization checks.

n	$C_{n,1}$	$C_{n,2}$	$C_{n,3}$	$C_{n,4}$
0	1	0	0	0
1	1	0	0	0
2	6	1	0	0
3	150	69	18	9
4	13032	10800	4992	4254
5	3513720	4339440	2626800	3015450
6	2722682160	4513642920	3177532800	4466769300

The $n = 6$ entries are exact transform computations checked against the OEIS comment; the exhaustive matrix audit described later stops at $n = 5$.

Remark 1.4 ([write] The four OEIS entries, quoted, and the case $k = 1, 6$ October 2026). The entries were read on 6 October 2026 in the OEIS internal format; the quotations are verbatim.

- (a) A089479 [1] (revision #31, 19 February 2026, the revision the source read) is named

`Triangle T(n,k) read by rows, where T(n,k) = number of times the permanent of a real n X n (0,1)-matrix takes the value k, for n >= 0, 0 <= k <= n!.`

so $C_{n,k} = T(n,k)$, and its row $n = 0$ reads 0, 1, the convention $\text{per}(\emptyset) = 1$ used here. Its comment gives the row $n = 6$ computed by Gordon Royle, whose entries for $k = 1, \dots, 4$ are the last line of the table above. The write recomputed the table independently, from (10)–(16) and the transform (23) in exact arithmetic with its own code: rows $0 \leq n \leq 6$ of the entry agree for $1 \leq k \leq 4$. The entry states no asymptotic formula.

- (b) A089482 [2] (revision #33, 19 February 2026), the column $k = 1$ (“`T(n,1) = A089482(n)`” in A089479), is named “`Number of real {0,1}-matrices having permanent = 1.`” and has the formula “`a(n) = n! * A003024(n).`” – `_Vladeta Jovovic_, Oct 26 2009`, with Max Alekseyev’s proof in its comments: the case $k = 1$ of (7).

- (c) A003024 [3] (revision #239, 13 August 2026) is named “Number of acyclic digraphs (or DAGs) with n labeled nodes.” and states

$a(n) \sim n! \cdot 2^{(n(n-1)/2)} / (M \cdot p^n)$, where $p = A245654 = 1.488078545599710294656246\dots$ is the root of the equation $\text{Sum}_{\{n \geq 0\}} (-1)^n p^n / (n! \cdot 2^{(n(n-1)/2)}) = 0$, and $M = \text{Sum}_{\{n \geq 1\}} (-1)^{(n+1)} p^n / ((n-1)! \cdot 2^{(n(n-1)/2)}) = A245655 = 0.57436237330931147691667\dots$ [...] - _Vaclav Kotesovec_, Dec 09 2013

(the omitted words record that two printings of the 2003–2004 paper of McKay et al. give $M = 0.474$, and Sloane’s reply that this is a typo for 0.574, “taken from Stanley’s 1973 paper”). *Theorem 1.1 at $k = 1$ is an instance of this classical asymptotic, not a new result.* Indeed $p = \rho$: the equation for p is $E(p) = 0$, and A245654 (revision #19, 20 September 2026) is the “smallest positive root” of that function [10], which is ρ by Lemma 5.1. And $M = a$: since $E'(z) = \sum_{m \geq 1} (-1)^m z^{m-1} / ((m-1)! 2^{\binom{m}{2}})$, the series for M is $-\rho E'(\rho)$ term by term, which is (1). With $P_1 = 1/a$ and (7), Theorem 1.1 reads $A003024(n) = n! 2^{\binom{n}{2}} a^{-1} \rho^{-n} \{1 + O((\rho/2)^n)\}$, the displayed formula with the relative error (35) at $d = 0$. The write’s 60-digit values of ρ and a (mpmath) lie in the enclosures of Section 7 and agree with the first 55 of the 103 digits printed in each of A245654 and A245655 [11] (revision #15, 21 January 2015; the further digits were not compared); so $M = 0.574\dots$, not 0.474. (The independent check of 6 October 2026 compared all 103 digits of each entry with 130-digit values of its own: they agree.)

- (d) The same entry has the comment

Also the number of $n \times n$ real $(0,1)$ -matrices with permanent equal to 1, up to permutation of rows/columns, cf. A089482. - _Vladeta Jovovic_, Oct 28 2009

which is right with “rows/columns” read as “rows, or columns”, and wrong if both are permuted at once. *Proof.* A matrix A of permanent one has pairwise distinct columns: if columns $i \neq j$ were equal, then exchanging i and j in its unique perfect matching would give a second one. So a nonidentity column permutation never fixes A , every orbit of the column action on the matrices counted by A089482 has exactly $n!$ elements, and there are $A089482(n)/n! = A003024(n)$ orbits; rows behave the same way, since $\text{per } A^T = \text{per } A$. Under rows and columns jointly, the six matrices of permanent one at $n = 2$ form two orbits (the identity with the antidiagonal matrix, and the four matrices with exactly three ones), while $A003024(2) = 3$. In general the joint orbits are the isomorphism classes of acyclic digraphs: permuting the rows by π and the columns arbitrarily replaces the digraph D of Section 2 (the unique matching moved to the diagonal) by $\pi(D)$, and conversely $I + A_{\pi(D)} = P_\pi(I + A_D)P_\pi^{-1}$; so their number is the number of unlabelled acyclic digraphs, OEIS A003087 [12] (1, 2, 6, 31 for $n = 1, \dots, 4$, as an exhaustive enumeration confirms). (Added after the independent check of 6 October 2026.) This is recorded here only; nothing was submitted to the OEIS.

- (e) A350790 [9] (revision #28, 18 April 2023), “Number of digraphs on n labeled nodes with a global source and sink.”, is cited by the source only because it links Robinson’s 1995 paper, “Counting digraphs with restrictions on the strong components” (Section 14); its sequence is not used.

1.2 Provenance, sources and reading conventions

[write] Added 6 October 2026, batch 108. *Provenance.* This report is Report 184 of the session bundle that ProveIt’s docs/incoming intake processed as batch 108: the archive FixedPermanentMatrices_

`Asymptotics_and_Inverses_Source.zip` (659,214 bytes, SHA-256 0863547a2a49...6957c3112e, 30 files at the archive root, no wrapper directory), arrival commit 60f54ea06, placed in 602e5bd0f. The text is the delivered `Report184.tex` (930 lines, dated 3 October 2026), shipped as `article.tex`; the delivered programs and data are shipped under `code/` and `data/` (the report README maps every name). The delivered 24-page PDF and the checksum manifest `SHA256SUMS.json` (29 entries, verified at placement) are not shipped, and the delivered README is replaced by the report README; all three are retrievable from the arrival commit. The package records no ProveIt commit; it describes a bounded look at report collections (Section 14) that is consistent with the repository but unpinned. It carries no “prepared for private review” line and no personal data. It is a single-source report, so no merge choice was made. The write prefixed the 83 delivered labels with `fpm:` and updated the 75 references to them; it added this subsection, the notes marked [write], Remarks 1.4 and 12.2, one question in Section 15 and two bibliography entries. The added remarks are the last statements of their sections, the added subsection follows the last delivered one of Section 1, and the added displays are unnumbered, so every theorem, section and equation of the source keeps its delivered number.

The sources, as the write read them (6 October 2026).

- OEIS A089479, A089482, A003024 and A350790, and the constants A245654 and A245655, in the internal format (Remark 1.4).
- De Panafieu and Dovgal [4] (arXiv:1903.09454v2, 7 pages): Theorem 3.4 is the transform (19) for digraphs whose strong components belong to a given family, proved by marking source-like components with u and setting $u = -1$, the argument the source repeats in Section 4; the authors credit Robinson’s 1995 paper for a similar proof. The source applies it with the components weighted in the finite-divisor algebra and gives that marked proof itself.
- Dovgal and Nurligareev [6] (arXiv:2310.05282v3, dated 5 August 2026): Section 4.3 (Corollary 4.14, Remark 4.15) gives the probability of $m + 1$ strongly connected components, dominated by one large component and m one-vertex components, as the source says; Section 4.4 treats several kinds of components.
- De Panafieu and Dovgal [5] (arXiv:2001.08659v2): the elementary digraphs (strong components isolated vertices or cycles), Section 2.3, and the regime $m = n(1 + \mu n^{-1/3})$ of edges, as the source says.
- Greenhill, Hasheminezhad, Iliffe and McKay [8] (arXiv:2601.04822v1): Theorems 4.2–4.5 concern the expected permanent under prescribed degrees, as the source says.
- The transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`: `p0:def:core`, `p0:thm:lambert-core`, `p0:prop:factorial-core`, `p0:def:three-inverses`, `p0:thm:staircase`, `plt:def:lw-monomial-log-datum` and `plt:thm:lw-template`, read for Remark 12.2; and, when the independent check was applied, `p0:thm:core-reversion`.
- Not read by the write: Kim, Lee and Seol [7] (the source read it in full); Robinson [9], whose host `cobweb.cs.uga.edu` refused connections on 6 October 2026 as during the source’s review.

What was checked at intake. The batch-108 dossier read the manuscript in full and found no error. It rechecked by hand the identity $a = -\rho E'(\rho) = \rho E(\rho/2)$ (since $E'(z) = -E(z/2)$), $S_3(1/2) = 1$, $S_4(1/2) = 91/6$, the sum 26 in (33), the three slack fractions of Section 8, the principal-part bounds 33, 60, 99, the lower bound for P_4 in Section 11.3 and the injection behind (44); and it reran the delivered suite on copies (note at the end of Section 13). The write rechecked every proof line by line and found no error. Independently of the delivered code it enumerated the kernels of Proposition 3.2 for $k = 3, 4$ (the counts 1, 2 and 1, 6, 24, 48 of the table in Section 3, and the rational functions (15) and (16), rebuilt from the kernel formula with SymPy); recomputed the table of Section 1 (Remark 1.4); extracted the thirteen quantities of Section 7 at 50 digits, the principal-part coefficients by contour integrals around ρ rather than by the derivative formulas (29)–(30) (all lie in their printed intervals); recomputed the slack of (49), the table of $\pi_2(m)$ and the rows $n = 5, 10$ of the discrepancy table of Section 13; and checked U_0, U_1, U_2 of Theorem 12.1 by series algebra (and $\deg U_3 = 4$, so the degree bound is attained at $j = 3$). Finite and floating computations are observations, not certificates.

Relation to the repository. Before batch 108 no file of the repository named A089479, A089482, A003024

or A350790, and no report counted binary matrices by their permanent (`a000382-winding-correction` evaluates permanents of particular 0–1 matrices, a different question). No statement of this report is formalized in Lean or Rocq, nothing in the repository’s formal developments concerns permanents or acyclic digraph counts, and the report’s place in the collection confers no formal status. The neighbouring reports (all under `SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/`) are `a182220-source-boundary`, whose extensional acyclic digraphs form a subclass of the digraphs counted at $k = 1$ and whose marked-source identity (its Proposition `sbd:prop:marked`) is the source-vertex form of the marked-source-component sieve proving (19); `a116379-bounded-identity-trees`, a different class and a different singularity (a square-root law), related by method only: rational interval certificates of the asymptotic constants (its Part II) and computable asymptotic inverses; and, see also, `a397711-bounded-indegree-dags` (batch 108), which counts labelled acyclic digraphs with bounded indegree, the unbounded count being A003024. The source’s review also checked `a299907-lonesum-decomposable-matrices`, `a372395-acyclic-orientation-partitions` and `a108242-regular-cyclic-word-covers` and found different models; that is so at the write. The shipped `code/verify_manifest.py` is byte for byte the helper `code/182-strict-verify_manifest.py` of `a271619-strict-twice-partitions`, a generic tool of the same bundle, shipped here again so that this package reruns on its own. The inverses of Sections 11–12 are compared with the transseries volume in Remark 12.2.

[write] What is not claimed, collected from the source. Nothing is claimed for a growing permanent $k = k(n)$, growing θ or growing expansion order; every constant may depend on k . The case $k = 1$ and the component transform are classical and credited; the source’s literature review is bounded and is “not a worldwide priority determination”, and Robinson’s 1995 paper was not obtained. Decimal diagnostics are not certificates, the C++ audits are finite, and finite comparisons prove no all- n statement; the all- n constants M_2, M_3, M_4 are deliberately safe, not optimized. No certified inverse solver is implemented, and the two-ceiling brackets do not yield an eventual exact rounding formula. No assertion that all zeros of E are real is made, and no convergent sum over all poles. The smooth envelope F_k is not an interpolation of the counts. Byte identity of rebuilt PDFs is promised only on the same Python/TeX stack. The write adds: its independent checks above are floating or finite; Remark 1.4(d) concerns the wording of an OEIS comment only; and Remark 12.2 claims no novelty for any inversion.

Reading conventions (letters the source reuses, with the tempting false readings; no symbol of the source was renamed; symbols of the transseries volume carry the subscript “vol” in Remark 12.2).

Symbol	Meaning here (where)	Not to be read as
a	$-\rho E'(\rho)$, (1); A003024’s M	matrix entries a_{ij} ; vertex counts a , b of Section 4
b, c	Taylor coefficients of E at ρ (Section 7); $c = d + 1$ in Section 12	$b_{n,k}$ (digraph counts); $c_{k,j}$, $c_{k,\zeta,j}$ (principal parts); the circulation c of Lemma 3.1
d, e, e_p, e_j	$\Omega(k)$; kernel excess $m - v$; prime exponents; basis of the divisor algebra	a degree or the number e
h_p, h_k, H_j	prime amplitudes (3); $h_k = H_k(\rho)$ in (29)	$H_4(\rho)$ as an amplitude (it is not)
L, L_k, ℓ	$z/(1 - z)$ (Section 3); leading coefficient (4); $\log 2$	each other
$M, M_k, \mathcal{M}_k$	kernel matrices; error constants of Theorem 1.2; block multisets	a multiplicity m
q, Q, Q_j	$\rho/2$ (Section 11); the cubic of Lemma 5.1; bundle factors (13)	each other
t, s, r	$1 - z/\rho$ (Section 6), $\sqrt{2Y/\ell}$ (Section 12); $\log t$; numbers of factors, $r_k(y)$ the smooth root	$s_{m,k}$; total degree t in Lemma 6.1
E, Δ, π_p, δ	the DAG series; damping (17); block-size law (6); $\log(2\pi L_k/k)$	an expectation; a difference; the number π ; point masses $\delta_{[\cdot]}$
$N_k(y)$	$\min\{n \geq 0 : C_{n,k} \geq y\}$	the staircase N_* of <code>p0: def: three-inverses</code> (equal for large y , Remark 12.2); the truncation $N = 45$

2 Marking one perfect matching

A matrix of permanent k has exactly k perfect matchings in its associated bipartite graph. Count a matrix together with one marked matching. Permute its columns by the unique permutation making that matching the main diagonal, and remove the diagonal entries. This gives a loopless simple directed graph D on $[n]$; an off-diagonal entry (i, j) is the arc $i \rightarrow j$.

Conversely, from any such digraph with $\text{per}(I + A_D) = k$ and any column permutation, one reconstructs a unique matrix with a marked matching. The mark is part of the object, so coincidences between unmarked matrices from different permutations cause no quotient. If

$$b_{n,k} = \#\{D \text{ on } [n] : \text{per}(I + A_D) = k\},$$

then

$$kC_{n,k} = n!b_{n,k}. \quad (7)$$

Every nonidentity permutation in $\text{per}(I + A_D)$ is a collection of pairwise vertex-disjoint directed simple cycles of lengths at least two; unselected vertices use their diagonal entry. The empty cycle collection is included. A directed cycle lies in one strongly connected component (SCC), so

$$\text{per}(I + A_D) = \prod_{B \text{ an SCC of } D} \text{per}(I + A_{D[B]}). \quad (8)$$

A singleton SCC has permanent one. Every nontrivial SCC contains a directed cycle and has permanent at least two.

Let $s_{m,k}$ count the strongly connected digraphs of the indicated kind, and set

$$S_k(z) = \sum_{m \geq 1} s_{m,k} \frac{z^m}{m!}. \quad (9)$$

Then $S_1(z) = z$, and $s_{m,k} = 0$ for $m < 2$ if $k \geq 2$. If $k = 1$, there can be no directed cycle at all. Thus $b_{n,1}$ is the number of labelled DAGs, the known correspondence recorded in A089482 and A003024 [2, 3]. The rest of the proof extends the component analysis to each fixed positive permanent.

3 Finite strongly connected kernels

3.1 A circulation bound

Lemma 3.1. *A strongly connected directed graph with v vertices and m arcs has at least $m - v + 1$ distinct directed simple cycles. Consequently, if its diagonal-one permanent is k , then $m - v \leq k - 2$.*

Proof. The directed incidence matrix has rank $v - 1$, so its real circulation space has dimension $m - v + 1$. Every arc lies on a directed cycle. Summing one such cycle circulation for each arc gives a strictly positive circulation c .

For any real signed circulation x , choose a sufficiently large positive t so that $x + tc$ is nonnegative. Every nonnegative circulation is a nonnegative linear combination of simple cycle circulations: start along a positive arc, continue along positive outgoing arcs using conservation at each vertex, stop at a repeated vertex, and subtract the minimum weight around the resulting simple cycle. At least one positive arc disappears in each subtraction. This terminates after finitely many steps. Thus all signed circulations lie in the span of the simple cycles, proving their number is at least the dimension of the space.

Every simple directed cycle is itself a distinct nonempty cycle collection in the permanent. Adding the empty collection gives $k \geq m - v + 2$. $\square$

When $k = 2$, strong connectivity gives $m \geq v$, and the lemma forces equality. Every indegree and outdegree is then one, so the graph is one directed cycle. Its labelled count is $(m - 1)!$, yielding

$$S_2(z) = \sum_{m \geq 2} \frac{z^m}{m} = -\log(1 - z) - z. \quad (10)$$

3.2 Unique suppression and permanent preservation

For $k \geq 3$, a strongly connected graph is not a single directed cycle. Call a vertex a branch vertex when its indegree and outdegree are not both one. Suppress all other vertices along their maximal directed paths. Each maximal path starts and ends at branch vertices: a cycle made entirely of suppressed vertices could not coexist with a branch vertex in a strongly connected graph. The resulting kernel is a directed multigraph K , with loops and parallel arcs allowed.

This suppression is unique. Each surviving vertex keeps its original indegree and outdegree, with loops counted once in each; no new suppressible branch vertex is created. Write $v = v(K)$, $m = m(K)$, and $e = m - v$. Subdivision and suppression preserve e . Every branch vertex has positive indegree and outdegree and total degree at least three. Therefore

$$3v \leq 2m = 2v + 2e, \quad v \leq 2e \leq 2k - 4, \quad m \leq 3k - 6. \quad (11)$$

Only finitely many kernels are possible for fixed k .

The permanent also survives suppression, with multiplicities correctly retained. A nonempty cycle collection using an internal vertex of a subdivided path must use the entire path, since each such vertex has a unique incoming and outgoing off-diagonal arc. It projects to a kernel cycle collection. Conversely every kernel cycle collection lifts uniquely. A loop at a branch vertex represents a genuine directed cycle through that vertex and at least one internal vertex. It competes with the identity choice at that vertex. Parallel arcs represent distinct possible paths and must be counted distinctly. If M_K is the integer adjacency matrix of the kernel, the invariant is exactly

$$\text{per}(I + M_K) = k. \quad (12)$$

There is no additional cycle collection wholly inside the interior of one suppressed path.

3.3 The exact finite kernel formula

Let $L = L(z) = z/(1 - z)$, the EGF of a nonempty ordered list of labelled internal vertices. Define

$$Q_0(z) = 1, \quad Q_j(z) = \frac{L^j}{j!} + \frac{L^{j-1}}{(j-1)!} \quad (j \geq 1). \quad (13)$$

For an off-diagonal bundle of j parallel kernel arcs, either every path is nonempty, or exactly one path is empty. Two empty paths would give the same original arc twice, which is forbidden. These two cases give (13). In a loop bundle every path must be nonempty, because the original digraph is loopless, so its factor is $L^j/j!$.

For each v , let $\mathcal{K}_{v,k}$ be the set of nonnegative integer $v \times v$ matrices M such that the associated directed multigraph is strongly connected, every indegree and outdegree is positive, none is simultaneously one, $\text{per}(I + M) = k$, and $\sum M_{ij} - v \leq k - 2$.

Proposition 3.2 (Finite kernel formula). *For $k \geq 3$,*

$$S_k(z) = \sum_{v=1}^{2k-4} \frac{z^v}{v!} \sum_{M \in \mathcal{K}_{v,k}} \prod_i \frac{L(z)^{M_{ii}}}{M_{ii}!} \prod_{i \neq j} Q_{M_{ij}}(z). \quad (14)$$

In particular, S_k is rational with no possible finite pole except $z = 1$.

Proof. The unique suppression associates one admissible kernel to each strongly connected graph of permanent k . The preceding bundle factors reconstruct exactly its simple lifts. An adjacency matrix M has branch positions labelled $1, \dots, v$; multiplying by $z^v/v!$ performs the labelled branch-vertex accounting. Dividing by each bundle factorial forgets permutations of indistinguishable kernel arcs. Every nonempty list carries distinct labels, so its internal labels distinguish the corresponding nonempty paths. There is at most one empty path in a nonloop bundle. Hence there is no additional arc-labelling factor.

Conversely, subdividing each kernel arc at least once always produces a simple strongly connected lift, and the permitted shorter choices produce exactly the other simple lifts. The permanent is preserved by (12). This proves the identity and the absence of hidden feasibility restrictions. Each summand is rational in L , and the sum is finite by (11). $\square$

The construction is effective, but it is not a claim that exhaustive kernel enumeration is practical for large k . Bounds (11) give a finite algorithm for every individual k .

3.4 The first rational component functions

For $k = 3$, the excess is at most one. The kernels are a one-vertex bouquet of two loops and an oriented theta with two parallel arcs in one direction and one in the other. There are two labelled orientations of the latter. Formula (14) gives

$$\begin{aligned} S_3(z) &= \frac{zL^2}{2} + z^2 \left(\frac{L^2}{2} + L \right) (1 + L) \\ &= \frac{z^3(3 - 2z)}{2(1 - z)^3}. \end{aligned} \quad (15)$$

For $k = 4$, enumerating the finite set in (14) gives

$$S_4(z) = \frac{z^3(-6z^4 + 28z^3 - 47z^2 + 28z + 6)}{6(1 - z)^6}. \quad (16)$$

The exact code checks the rational numerator, without numerical fitting. The numbers of labelled kernels by branch count are

k	$v = 1$	$v = 2$	$v = 3$	$v = 4$
3	1	2	0	0
4	1	6	24	48

All listed $k = 4$ kernels have excess two. The finite formula, rather than a guess from initial coefficients, certifies (16).

For every $k \geq 2$ there is a strongly connected example: take $k - 1$ directed 2-cycles sharing one common vertex and otherwise disjoint. Every nonempty cycle collection consists of exactly one of these cycles. Its permanent is k . This also proves that every prime amplitude used below is nonempty.

4 The classical component transform with a permanent mark

Define the coefficient-damping operator

$$\Delta \left(\sum_{m \geq 0} f_m z^m \right) = \sum_{m \geq 0} \frac{f_m z^m}{2^{\binom{m}{2}}}. \quad (17)$$

It converts an EGF to the graphic generating function when their coefficients are expressed consistently. The relevant series is

$$D_k(z) = \sum_{n \geq 0} \frac{b_{n,k}}{n! 2^{\binom{n}{2}}} z^n = \sum_{n \geq 0} \frac{k C_{n,k}}{(n!)^2 2^{\binom{n}{2}}} z^n. \quad (18)$$

To justify all operations without global convergence in the permanent index, fix k and use the finite-divisor algebra with basis e_j for $j \mid k$, unit e_1 , and multiplication

$$e_i e_j = \begin{cases} e_{ij}, & ij \mid k, \\ 0, & ij \nmid k. \end{cases}$$

This multiplication is associative: once a product fails to divide k , multiplying it by another positive integer cannot make it divide k . The ideal generated by e_j , $j \geq 2$, is nilpotent of index at most $d + 1$. Every coefficient identity below can be read in this fixed algebra.

Set

$$\mathcal{S}(z) = ze_1 + \sum_{\substack{j|k \\ j \geq 2}} S_j(z)e_j, \quad \mathcal{D}(z) = \sum_{j|k} D_j(z)e_j.$$

The classical SCC identity of Robinson, restated with proof in Theorem 3.4 of de Panafieu and Dovgal [4], is

$$\mathcal{D}(z) = [\Delta \exp(-\mathcal{S}(z))]^{-1}. \quad (19)$$

Its multiplicative permanent mark is justified by (8).

For completeness, mark an arbitrary subset of the source SCCs of a digraph and give each selected SCC weight -1 . Distinct source SCCs have no arcs between them. Their labelled collection is therefore a SET, giving $\exp(-\mathcal{S})$ before applying Δ . Arcs from the selected vertices to the remaining digraph may be chosen arbitrarily, while reverse arcs are forbidden. If the two sides have a and b vertices, there are 2^{ab} choices. This arrow product becomes ordinary multiplication in graphic normalization, because

$$\binom{a+b}{2} = \binom{a}{2} + \binom{b}{2} + ab.$$

Every nonempty condensation DAG has a source. Summing the signs over its possible selected source subsets yields zero, while the empty graph contributes one. This proves (19) in the marked algebra. It also explains why the denominator is $2^{\binom{n}{2}}$ even though both directions of each possible arc are allowed in the underlying digraph class.

Write

$$\Delta \exp(-\mathcal{S}(z)) = E(z)e_1 - \sum_{\substack{j|k \\ j \geq 2}} H_j(z)e_j. \quad (20)$$

Explicitly, for $j \geq 2$,

$$H_j(z) = \sum_{r=1}^{\Omega(j)} \frac{(-1)^{r+1}}{r!} \sum_{\substack{j_1 \cdots j_r = j \\ j_i \geq 2}} \Delta \left(e^{-z} \prod_{i=1}^r S_{j_i}(z) \right). \quad (21)$$

Every H_j is entire. Indeed, every fixed product on the right is analytic for $|z| < 1$ and has coefficients bounded by $C_R R^{-m}$ on each fixed circle of radius $R < 1$. The damping factor $2^{-\binom{m}{2}}$ makes the transformed series converge absolutely and uniformly on every compact complex disk. All such statements concern a fixed finite-divisor algebra, not the unrestricted sum over all permanents.

Expanding the inverse using its nilpotent ideal gives the exact finite expression

$$D_1 = \frac{1}{E}, \quad D_k = \sum_{r=1}^d \frac{1}{E^{r+1}} \sum_{\substack{j_1 \cdots j_r = k \\ j_i \geq 2}} \prod_{i=1}^r H_{j_i} \quad (k > 1). \quad (22)$$

Thus every D_k is meromorphic on the whole complex plane and its only possible poles are zeros of E .

For later computation the first cases are

$$\begin{aligned}
D_2 &= H_2/E^2, & H_2 &= \Delta(e^{-z}S_2), \\
D_3 &= H_3/E^2, & H_3 &= \Delta(e^{-z}S_3), \\
D_4 &= H_4/E^2 + H_2^2/E^3, & H_4 &= \Delta\{e^{-z}(S_4 - S_2^2/2)\}.
\end{aligned} \tag{23}$$

The subtraction in H_4 is necessary. Treating it as simply $\Delta(e^{-z}S_4)$ would lose the multiplicative component accounting and give an incorrect polynomial.

5 A self contained complex root isolation

Lemma 5.1. *E has exactly one zero in $|z| \leq 2$. It is simple and real, lies in $(37/25, 3/2)$, and $E(x) > 0$ for $0 \leq x < \rho$. On $|z| = 2$,*

$$|E(z)| > \eta := 119/1000. \tag{24}$$

Proof. Let

$$Q(z) = 1 - z + z^2/4 - z^3/48.$$

Its derivative is $Q'(z) = -(z - 4)^2/16$, so it has exactly one real root r_0 . Direct evaluation puts $r_0 \in (1.48, 1.50)$. The other two roots are conjugate nonreal numbers with common modulus $\sqrt{48/r_0} > \sqrt{32}$, using the product of the three roots. Hence on $|z| = 2$,

$$|Q(z)| > \frac{1}{48} \cdot \frac{1}{2} \cdot (\sqrt{32} - 2)^2 > \frac{13}{100}.$$

The first term of $E - Q$ on that circle has absolute value $1/96$. The ratio of consecutive absolute terms is at most $1/40$ from index four onward. Therefore

$$|E(z) - Q(z)| \leq \frac{1/96}{1 - 1/40} = \frac{5}{468} < \frac{11}{1000}.$$

Rouché's theorem gives exactly one zero, counted with multiplicity, inside the circle, and the strict bounds exclude boundary zeros. Conjugation makes that zero real and the count makes it simple. Alternating-series bounds give explicit real signs:

$$E(37/25) > Q(37/25) = \frac{47}{750000} > 0,$$

$$E(3/2) < Q(3/2) + \frac{(3/2)^4}{1536} = -\frac{37}{8192} < 0.$$

This isolates the root. Since $E(0) = 1$ and there is no earlier real zero, E is positive on $[0, \rho)$. Subtracting the two circle bounds gives (24). $\square$

No assertion about all zeros of E being real is required. Its other zeros enter only through their actual locations and multiplicities when a larger finite radius is chosen in Section 9.

6 Positive prime amplitudes and the exact pole degree

Alternating transforms do not preserve positivity in general. The needed positivity has a special direct proof.

Lemma 6.1. *If $S(z) = \sum_{m \geq 2} s_m z^m / m!$ has radius of convergence at least one, then*

$$\Delta(e^{-z}S)(z) = \sum_{m \geq 2} \frac{s_m z^m}{m! 2^{\binom{m}{2}}} E(z/2^m). \quad (25)$$

Both sides are entire and the displayed sum converges locally uniformly.

Proof. Expand e^{-z} and write the total degree as $m + t$. The identity

$$\binom{m+t}{2} = \binom{m}{2} + \binom{t}{2} + mt$$

turns the inner sum over t into $E(z/2^m)$, with the factors $m!$ and $t!$ unchanged. Absolute local convergence follows by bounding $s_m/m!$ by an exponential on a smaller initial disk and using the quadratic damping. Thus regrouping is valid. $\square$

For prime p , the only factorization in (21) has one factor, so $H_p = \Delta(e^{-z}S_p)$. At $z = \rho$, every $E(\rho/2^m)$ is strictly positive by Lemma 5.1; every $s_{m,p}$ is nonnegative; and at least one is positive by the bouquet construction. Hence $H_p(\rho) = h_p > 0$, proving (3).

Let $t = 1 - z/\rho$. Near ρ ,

$$E(z) = at + O(t^2), \quad a > 0.$$

The greatest number of factors in an ordered factorization of k is $d = \Omega(k)$, and equality occurs only for its ordered prime factorizations. Consequently the coefficient of the highest denominator power E^{-d-1} in (22) is

$$\frac{d!}{\prod_p e_p!} \prod_p H_p(z)^{e_p}. \quad (26)$$

At ρ this is strictly positive; terms with fewer factors have strictly lower pole order. Thus D_k has a pole of exact order $d + 1$.

Write its complete principal part as

$$\text{PP}_\rho D_k(z) = \sum_{j=1}^{d+1} c_{k,j} (1 - z/\rho)^{-j}, \quad P_k(n) = \sum_{j=1}^{d+1} c_{k,j} \binom{n+j-1}{j-1}. \quad (27)$$

The polynomial has real coefficients, since ρ and the series are real. The top coefficient in (26), divided by a^{d+1} and then by $d!$ from coefficient extraction, gives exactly (4).

Subtracting (27) leaves a function analytic on a neighborhood of the closed disk $|z| \leq 2$, by Lemma 5.1 and meromorphicity. Cauchy's coefficient bound on that circle gives

$$[z^n]D_k(z) = \rho^{-n} P_k(n) + O_k(2^{-n}).$$

Combining this with (18) proves Theorem 1.1. This argument also covers $k = 1$ directly with $P_1 = 1/a$.

7 The first complete polynomial prefactors

The binomial basis in (27) is the natural basis for the answer. In particular,

$$\begin{aligned} P_2(n) &= c_{2,2}(n+1) + c_{2,1}, \\ P_3(n) &= c_{3,2}(n+1) + c_{3,1}, \\ P_4(n) &= c_{4,3}(n+1)(n+2)/2 + c_{4,2}(n+1) + c_{4,1}. \end{aligned} \tag{28}$$

The leading coefficient L_4 is $c_{4,3}/2$, not $c_{4,3}$.

7.1 Exact derivative formulas

Write, at $z = \rho$,

$$b = \rho^2 E''(\rho)/2, \quad c = -\rho^3 E'''(\rho)/6, \quad E(z) = at + bt^2 + ct^3 + O(t^4).$$

For $k = 2, 3$, writing $h_k = H_k(\rho)$ and $h'_k = H'_k(\rho)$ gives

$$c_{k,2} = \frac{h_k}{a^2}, \quad c_{k,1} = -\frac{\rho h'_k}{a^2} - \frac{2bh_k}{a^3}. \tag{29}$$

For $k = 4$, abbreviate $h = H_2(\rho)$, $h' = H'_2(\rho)$, $h'' = H''_2(\rho)$, $g = H_4(\rho)$, and $g' = H'_4(\rho)$. Then

$$\begin{aligned} c_{4,3} &= h^2/a^3, \\ c_{4,2} &= g/a^2 - 2\rho h h'/a^3 - 3bh^2/a^4, \\ c_{4,1} &= -\rho g'/a^2 - 2bg/a^3 + \rho^2\{(h')^2 + hh''\}/a^3 \\ &\quad + 6b\rho h h'/a^4 + h^2(6b^2/a^5 - 3c/a^4). \end{aligned} \tag{30}$$

These follow by multiplying the Taylor series of (23) in t . For example,

$$H_2(z)^2 = h^2 - 2\rho h h' t + \rho^2\{(h')^2 + hh''\}t^2 + O(t^3),$$

which fixes both the second-derivative and cross-term factors.

7.2 Outward rational decimal intervals

Every endpoint in the following table is an exact terminating decimal. Each listed quantity lies between the two endpoints, and every interval has width 10^{-25} . These are certified enclosures, not floating-point error estimates.

Quantity	Lower endpoint	Upper endpoint
ρ	1.4880785455997102946562460	1.4880785455997102946562461
a	0.5743623733093114769166708	0.5743623733093114769166709
h_2	0.4979712937691307225327241	0.4979712937691307225327242
h_3	0.7957739216056920566378634	0.7957739216056920566378635
$H_4(\rho)$	1.3898313409494419980287943	1.3898313409494419980287944
$c_{1,1}$	1.7410611252932298403421880	1.7410611252932298403421881
$c_{2,2}$	1.5094973162987879512533343	1.5094973162987879512533344
$c_{2,1}$	-4.8156204403511959734926939	-4.8156204403511959734926938
$c_{3,2}$	2.4122245881933563368810739	2.4122245881933563368810740
$c_{3,1}$	-10.9317703274957438628613269	-10.9317703274957438628613268
$c_{4,3}$	1.3087318502556789012954150	1.3087318502556789012954151
$c_{4,2}$	-3.3028358986717741260484241	-3.3028358986717741260484240
$c_{4,1}$	-2.3699604764464618517018264	-2.3699604764464618517018263

Here $H_4(\rho)$ is a convenient numerator value; it is not a prime amplitude in formula (4).

7.3 Proof of the numerical certificate

All certificate arithmetic is integer or rational. The script retains terms of degrees 0 through 44 and uses the exact root bracket

$$\begin{aligned}\rho_- &= 1.4880785455997102946562460315823576618935, \\ \rho_+ &= 1.4880785455997102946562460315823576618936.\end{aligned}$$

Let $N = 45$ and $R = 3/2$. For $0 \leq j \leq 4$, the first omitted absolute term of $E^{(j)}$ on $|z| \leq R$ is bounded by

$$A_{E,j} = \frac{R^{N-j}}{(N-j)!2^{\binom{N}{2}}}. \quad (31)$$

The consecutive ratio is $R/((n+1-j)2^n) < 1/2$ for $n \geq N$. Hence twice (31) bounds the whole tail.

For the H functions, use $W_2 = S_2$, $W_3 = S_3$, and $W_4 = S_4 - S_2^2/2$. On $|z| = 1/2$, nonnegative coefficients of S_j and (10)–(16) give

$$S_2(1/2) < 1/4, \quad S_3(1/2) = 1, \quad S_4(1/2) = 91/6.$$

The first inequality follows termwise from $1/m \leq 1/2$ for $m \geq 2$, strictly for some terms. Also $e^{1/2} < 2$, by the exponential series comparison with a geometric series. Thus all three functions $e^{-z}W_k$ have modulus below 64 on this circle; for the largest case the bound is already

$$2(91/6 + 1/32) = 1459/48 < 64.$$

Cauchy's coefficient bound gives $|[z^m]e^{-z}W_k| \leq 64 \cdot 2^m$. After graphic damping, the first omitted absolute term of $H_k^{(j)}$ is at most

$$A_{H,j} = \frac{64 \cdot 2^N (N)_j R^{N-j}}{2^{\binom{N}{2}}}, \quad (N)_j = N(N-1) \cdots (N-j+1). \quad (32)$$

Its consecutive ratio for $n \geq 45$ is

$$\frac{2R(n+1)}{(n+1-j)2^n} < \frac{1}{2}.$$

Again twice the first term is a valid tail bound.

The coefficients through degree 44 are exact fractions obtained from the finite kernel formula and $[z^m]S_2 = 1/m$. Interval Horner evaluation with the above tails proves $E(\rho_-) > 0$ and $E(\rho_+) < 0$. Lemma 5.1 makes this the desired unique root. The same interval arithmetic evaluates the derivatives and formulas (29)–(30); its positive enclosure for a excludes division by zero. Finally the lower endpoint is rounded down and the upper endpoint up by exact integer floor and ceiling at denominator 10^{25} .

The certificate guards the coefficient length, derivative order $0 \leq j \leq 4$, derivative type, and disk $|z| \leq 3/2$. These are part of the proved domain, not optional assumptions. A second independent rational implementation using 50 terms and formal Laurent-series inversion reproduces the displayed intervals. The bundled primary certificate is sufficient to replay their proof.

[write] The second implementation is not shipped, 6 October 2026. The “second independent rational implementation using 50 terms and formal Laurent-series inversion” named in the preceding paragraph is not part of the delivered package: none of its 30 files implements it, and neither README lists it. The closest shipped program, `code/decimal_diagnostics.py`, is a floating Decimal computation from 70 Taylor coefficients and the derivative formulas, which the delivered README calls non-certified; it is not that

implementation. Its agreement with the table therefore cannot be checked from the package, and the claim is recorded as a question in Section 15 (Question 7). The enclosures do not depend on it: they are the output of the shipped `code/certify_constants.py`, which reproduces `data/constant_certificate.json` byte for byte in normal and optimized Python (rerun note at the end of Section 13). As a further, numerical, cross-check the write computed the thirteen quantities with its own code at 50 significant digits (mpmath), the principal-part coefficients $c_{k,j}$ by contour integrals of $D_k(\rho(1-t))t^{d+1}$ on $|t| = 1/20$ rather than through (29)–(30); every value lies in its printed interval. That is a floating-point computation, not a second certificate.

8 Explicit all index error constants

This section proves Theorem 1.2. The constants are deliberately safe, rather than optimized.

The circle bound (24) gives $|E| > \eta = 119/1000$ on $|z| = 2$. The preceding $|z| = 1/2$ estimates can be sharpened individually to

$$\begin{aligned} |[z^m]e^{-z}S_2| &< \tfrac{1}{2} 2^m, \\ |[z^m]e^{-z}S_3| &< 2 \cdot 2^m, \\ |[z^m]e^{-z}(S_4 - S_2^2/2)| &< 32 \cdot 2^m. \end{aligned}$$

The last follows from $1459/48 < 32$. Also

$$\sum_{m \geq 0} \frac{4^m}{2^{\binom{m}{2}}} < 27. \quad (33)$$

Indeed the terms through $m = 5$ sum to 26; the next is $1/8$ and subsequent ratios are at most $1/16$, so the remainder is at most $2/15$.

It follows that on $|z| = 2$,

$$|H_2| < 27/2, \quad |H_3| < 54, \quad |H_4| < 864.$$

Using (23) gives

$$\begin{aligned} |D_2| &< (27/2)\eta^{-2}, \\ |D_3| &< 54\eta^{-2}, \\ |D_4| &< 864\eta^{-2} + (27/2)^2\eta^{-3}. \end{aligned}$$

Since $\rho < 3/2$,

$$|1 - z/\rho|^{-1} \leq \rho/(2 - \rho) < 3 \quad (|z| = 2).$$

The certified intervals imply

$$\begin{aligned} |c_{2,1}| &< 5, \quad |c_{2,2}| < 2, \quad |c_{3,1}| < 11, \quad |c_{3,2}| < 3, \\ |c_{4,1}| &< 3, \quad |c_{4,2}| < 4, \quad |c_{4,3}| < 2. \end{aligned}$$

Thus the absolute bounds for the complete principal parts on that circle are, respectively, 33, 60, and 99. Removing them leaves analytic remainders bounded by

$$\begin{aligned} (27/2)\eta^{-2} + 33 &< 1000, \\ 54\eta^{-2} + 60 &< 3900, \\ 864\eta^{-2} + (27/2)^2\eta^{-3} + 99 &< 170000. \end{aligned} \quad (34)$$

These are exact rational inequalities. Their positive slacks are

$$\frac{193687}{14161}, \quad \frac{378240}{14161}, \quad \frac{1244199259}{1685159}.$$

Cauchy's coefficient bound now proves (5) for every $n \geq 0$, including initial indices for which the polynomial contribution is negative or the matrix family is empty.

9 Every fixed asymptotic order

9.1 Factorials retained and Stirling expanded

Theorem 1.1 already gives every algebraic correction with factorials retained. Namely, if

$$P_k(n) = L_k n^d \left(1 + \sum_{j=1}^d p_{k,j} n^{-j} \right),$$

then the finite expression on the right is exact as a polynomial. The remaining relative error is

$$O_k((\rho/2)^n n^{-d}). \quad (35)$$

For n large, the positivity of L_k ensures the division used here is valid.

To remove the factorials, put

$$\ell = \log 2, \quad \beta = 2 + \log \rho + \ell/2. \quad (36)$$

The usual fixed-order Stirling expansion yields, for every fixed $J \geq 0$,

$$\begin{aligned} C_{n,k} &= \frac{2\pi L_k}{k} n^{d+1} \exp \left(\frac{\ell n^2}{2} + 2n \log n - \beta n \right) \\ &\quad \times \left(1 + \sum_{j=1}^J q_{k,j} n^{-j} + O_{k,J}(n^{-J-1}) \right). \end{aligned} \quad (37)$$

Each $q_{k,j}$ is effective: multiply $P_k/(L_k n^d)$ by the exponential of the squared-factorial logarithmic Stirling series

$$2 \sum_{r \geq 1} \frac{B_{2r}}{2r(2r-1)n^{2r-1}} = \frac{1}{6n} - \frac{1}{180n^3} + \frac{1}{630n^5} - \cdots. \quad (38)$$

This is a fixed-order Poincaré expansion, not a claim of convergence of the infinite series. For example, with missing p coefficients understood as zero,

$$q_{k,1} = p_{k,1} + 1/6, \quad q_{k,2} = p_{k,2} + p_{k,1}/6 + 1/72.$$

Finite Stirling remainders and the exponentially smaller error (35) establish (37) at each requested order.

9.2 Finite spectral sectors

Theorem 9.1 (Finite radius spectral expansion). *Fix k and $R > \rho$ such that E has no zero on $|z| = R$. For each zero ζ of E inside that circle, let*

$$\text{PP}_\zeta D_k(z) = \sum_{j=1}^{r_{k,\zeta}} c_{k,\zeta,j} (1 - z/\zeta)^{-j}, \quad P_{k,\zeta}(n) = \sum_j c_{k,\zeta,j} \binom{n+j-1}{j-1},$$

with the polynomial zero if the apparent pole cancels. Then

$$\frac{kC_{n,k}}{(n!)^2 2^{\binom{n}{2}}} = \sum_{\substack{E(\zeta)=0 \\ |\zeta| < R}} \zeta^{-n} P_{k,\zeta}(n) + O_{k,R}(R^{-n}). \quad (39)$$

If ζ has multiplicity m as a zero of E , then $r_{k,\zeta} \leq m(d+1)$.

Proof. Formula (22) gives the possible pole-order bound. There are only finitely many zeros inside the closed disk, since E is entire and not identically zero. Subtract all actual principal parts. The result is analytic through the circle, and Cauchy's bound gives (39). Conjugate poles pair to give real coefficients when needed. $\square$

This is the standard meromorphic coefficient-transfer argument. It allows cancellations, multiple zeros and nonreal zeros. No convergent sum over all poles is asserted, and the radius-two dominant sector requires none of these further zeros to be located numerically.

10 Canonical blocks and their probability laws

10.1 Why the blocks do not depend on the marked matching

Call an edge of the original bipartite graph *allowed* if it belongs to at least one perfect matching. Fix one perfect matching and normalize it onto the diagonal. An off-diagonal arc $i \rightarrow j$ is allowed if and only if it lies on a directed cycle. One direction follows by rotating the diagonal matching around that cycle. In the other direction, any alternative matching decomposes relative to the diagonal matching into vertex-disjoint permutation cycles.

An arc lies on a directed cycle if and only if its ends belong to the same SCC. Consequently the allowed bipartite graph has no edges between distinct SCC row/column blocks. Within an SCC, the diagonal matching together with all the directed arcs forms a connected bipartite graph: contracting the diagonal edges yields the connected underlying graph of that SCC. Thus the SCC blocks are exactly the connected components of the allowed-edge graph.

This is a matching-independent construction. Each component has equal numbers of row and column vertices because the fixed matching restricts to it. Its size and the permanent of its square submatrix are intrinsic. A singleton SCC corresponds to one allowed bipartite edge, with block permanent one. Entries between different components may exist in the original matrix, but none belongs to a perfect matching.

If a uniform matrix of permanent k is followed by a uniform choice of one of its k matchings, every marked pair is uniform. The bijection of Section 2 gives a uniform digraph counted by $b_{n,k}$, since each one has $n!$ marked preimages. There is no block-dependent sampling bias. Therefore SCC statistics established for these digraphs apply to the canonical blocks of uniform unmarked matrices.

10.2 The limiting size distribution

Let X_n be the multiset of pairs (j, m) , one for each nontrivial canonical block, where j is its permanent and m its row count. Its permanent labels multiply to k . Let $\mathcal{M}_k$ be the countable set of all such finite multisets, including the empty multiset for $k = 1$. For $X \in \mathcal{M}_k$, set

$$V(X) = \sum_{(j,m) \in X} m.$$

With $w_{p,m}$ denoting the numerator in (6), formula (3) proves $h_p = \sum_m w_{p,m} > 0$, so π_p is a probability law. Since S_p has radius at least one and $2^{-\binom{m}{2}}$ dominates every exponential, all its positive exponential moments are finite.

Let X_∞ be obtained by independently drawing e_p sizes from π_p for each prime $p \mid k$, marking each with permanent p , and then forgetting any order among the blocks. Repeated primes mean independent draws from the same law, followed by formation of an unordered multiset. No additional factorial belongs in its probability normalization.

Theorem 10.1 (Weighted expansions for the complete multiset). *For every fixed $k \geq 1$, $\theta \geq 0$ and integer $J \geq 0$, there are effectively computable signed measures $\nu_{k,0}, \dots, \nu_{k,J}$ on $\mathcal{M}_k$ such that $\nu_{k,0} = \mathcal{L}(X_\infty)$ and*

$$\left\| \mathcal{L}(X_n) - \sum_{r=0}^J n^{-r} \nu_{k,r} \right\|_\theta = O_{k,\theta,J}(n^{-J-1}), \quad \|\nu\|_\theta = \sum_{X \in \mathcal{M}_k} e^{\theta V(X)} |\nu(X)|. \quad (40)$$

For $r \geq 1$, the total mass of $\nu_{k,r}$ is zero. At $\theta = 0$ this norm bounds ordinary total variation, whichever of the usual factor-of-two conventions is used.

Proof. Use weighted ℓ^1 measures on all block multisets whose permanent products divide the fixed k . Multiply point masses by multiset union, setting the product to zero if its permanent product no longer divides k . The empty multiset is the unit. The norm is submultiplicative because V is additive and discarded products can only reduce the bound. This is a commutative unital Banach algebra. Its ideal of nonempty multisets is nilpotent of index at most $d + 1$.

Replace the scalar series S_j in the component transform by

$$S_j(z) = \sum_{m \geq 2} \frac{s_{m,j} z^m}{m!} \delta_{[(j,m)]}.$$

For any fixed θ , choose $R_0 < e^{-\theta}$. The norm of this series is bounded on $|z| = R_0$, because it is bounded by $S_j(e^\theta R_0)$. Products of at most d such series times e^{-z} have coefficient norms bounded by $C R_0^{-m}$. Applying Δ therefore produces entire Banach-valued functions, with norm convergence on every compact disk. The exact finite formulas (21) and (22) hold unchanged for the corresponding bold symbols.

For prime p , the coefficientwise form of (25) gives

$$H_p(\rho) = \sum_m w_{p,m} \delta_{[(p,m)]},$$

a positive measure of mass h_p . The leading principal coefficient of D_k at ρ is

$$\frac{d!}{a^{d+1} \prod_p e_p!} {}^* \!_p H_p(\rho)^{*e_p},$$

where $*$ denotes convolution by multiset union. Subtracting the full Banach-valued principal part and applying the radius-two Cauchy estimate gives

$$[z^n] \mathbf{D}_k(z) = \rho^{-n} \mathbf{P}_k(n) + O_{k,\theta}(2^{-n}) \quad (41)$$

in the weighted norm, for a measure-valued polynomial of degree at most d . The total-mass map gives exactly the scalar $P_k(n)$.

Divide (41) by the positive scalar coefficient $[z^n] \mathbf{D}_k$. The uniform marked-matrix normalization cancels. The leading coefficient of the measure polynomial divided by L_k is

$$*_p(\mathbf{H}_p(\rho)/h_p)^{*e_p},$$

which is precisely $\mathcal{L}(X_\infty)$. Expanding the reciprocal of $P_k(n)/(L_k n^d) = 1 + O_k(n^{-1})$ to any fixed order yields (40). The normalized exponential error is smaller than every fixed power n^{-1} . Total mass one on the left forces mass zero for every higher coefficient, or this follows directly by scalar formal division. $\square$

The leading law is supported on exactly d prime-permanent blocks. There can never be more than d nontrivial blocks, and equality holds exactly when every block permanent is prime. Taking $J = 0$ in (40) proves Theorem 1.3 and the $O_k(n^{-1})$ exceptional probability. Fixed exponential moments and all fixed polynomial moments of $V(X_n)$ have corresponding expansions. No uniformity in a growing k , growing θ , or growing truncation order is being claimed.

For $k = 1$, X_n is always empty. For prime k , every nonempty finite matrix family already has exactly one nontrivial block. In particular, for $k = 2$ that block is exactly a directed cycle. Its limiting probabilities begin

Cycle size m	Diagnostic $\pi_2(m)$
2	0.7354112130140739
3	0.2267783070809675
4	0.0349695725447146
5	0.0027303593128658
6	0.0001083409118175

These rounded probabilities are numerical illustrations. The exact law is (6), and its proof does not rely on the displayed digits.

11 Smooth envelopes and integer threshold inverses

For sufficiently large real x , $P_k(x) > 0$. Define

$$F_k(x) = \frac{\Gamma(x+1)^2}{k} 2^{x(x-1)/2} \rho^{-x} P_k(x), \quad q = \rho/2 < 1. \quad (42)$$

This is a target-specific smooth envelope, not an assertion of an exact canonical interpolation of the integer counts. Theorem 1.1 gives

$$C_{n,k}/F_k(n) = 1 + O_k(q^n n^{-d}).$$

Its logarithmic derivative satisfies

$$\begin{aligned} (\log F_k)'(x) &= 2\psi(x+1) + (x-1/2)\ell - \log \rho + P'_k(x)/P_k(x) \\ &= \ell x + 2\log x + O_k(1), \end{aligned} \quad (43)$$

so it is positive and bounded below by a positive multiple of x eventually. Write $r_k(y) = F_k^{-1}(y)$ on this final interval.

11.1 Monotonicity of the integer sequence

From any matrix counted by $C_{n,k}$ form

$$\begin{pmatrix} A & 0 \\ v & 1 \end{pmatrix}, \quad v \in \{0, 1\}^{1 \times n}.$$

The final column forces the final row's matched entry, so the permanent remains k . Distinct pairs (A, v) give distinct matrices. Hence

$$C_{n+1,k} \geq 2^n C_{n,k} \quad (n \geq 0). \quad (44)$$

The bouquet construction and singleton extensions guarantee eventual nonemptiness for every k . Thus the sequence is eventually strictly increasing and unbounded. More precisely it is strictly increasing for $n \geq 1$ whenever $C_{n,k} > 0$; the initial equality $C_{0,1} = C_{1,1} = 1$ is an exception to strictness. Define the global threshold

$$N_k(y) = \min\{n \geq 0 : C_{n,k} \geq y\}. \quad (45)$$

11.2 The two ceiling bounds

For a sufficiently large fixed $B > 0$ put

$$F_{k,\pm}(x) = F_k(x)(1 \pm Bq^x x^{-d})$$

on a final interval where they are positive and increasing. Their integer values sandwich $C_{n,k}$ there. After excluding the finite prefix by taking y sufficiently large, monotonicity gives

$$\left\lceil F_{k,+}^{-1}(y) \right\rceil \leq N_k(y) \leq \left\lceil F_{k,-}^{-1}(y) \right\rceil. \quad (46)$$

The upper counting envelope has the smaller inverse, which fixes the directions of the inequalities.

Both perturbed inverse roots lie within distance one of $r = r_k(y)$ eventually: their relative perturbations vanish while (43) grows linearly. On this unit neighborhood, $q^x x^{-d}$ is comparable with $q^r r^{-d}$ and the logarithmic derivative is bounded below by a positive multiple of r . The mean value theorem applied to the logarithms then gives

$$F_{k,\pm}^{-1}(y) = r_k(y) + O_k(q^{r_k(y)} r_k(y)^{-d-1}).$$

Consequently, for some fixed $B' > 0$ and every sufficiently large y ,

$$\left\lceil r_k(y) - B' q^{r_k(y)} r_k(y)^{-d-1} \right\rceil \leq N_k(y) \leq \left\lceil r_k(y) + B' q^{r_k(y)} r_k(y)^{-d-1} \right\rceil. \quad (47)$$

This is a pointwise statement, including values of y arbitrarily close to a threshold. It deliberately keeps the uncertainty inside two ceilings. Exponentially small smooth-root uncertainty alone does not imply one universally valid eventual rounding formula: the root may be equally close to an integer.

11.3 Explicit low permanent envelopes

For $k = 2, 3, 4$ define, for $x \geq 50$,

$$G_k(x) = \frac{\Gamma(x+1)^2}{k} 2^{x(x-1)/2} \rho^{-x}, \quad F_{k,\pm}^{\text{eff}}(x) = G_k(x) \{P_k(x) \pm M_k q^x\}. \quad (48)$$

The all- n estimate (5) sandwiches every integer value for $n \geq 50$. We next prove that both envelopes are positive and strictly increasing throughout the entire real interval.

The certified binomial-basis intervals imply, for $x \geq 50$,

$$\begin{aligned} P_2(x) &> 1.5x - 3.5, & P_2'(x) &> 1.5, \\ P_3(x) &> 2.4x - 8.6, & P_3'(x) &> 2.4, \\ P_4(x) &> 0.65x^2 - 1.45x - 4.5, & P_4'(x) &> 1.3x - 1.45. \end{aligned}$$

For the last line use $c_{4,3} > 1.3$, $c_{4,2} > -3.4$, and $c_{4,1} > -2.4$ in (28); the derivative multipliers are positive on this interval. Since $1/2 < q < 3/4$,

$$M_k q^x \leq 170000(3/4)^{50} < 1/10, \quad |(M_k q^x)'| < 1/10. \quad (49)$$

The first inequality is exact rational arithmetic, with slack

$$\frac{1}{10} - 170000(3/4)^{50} = \frac{1475750661002500018352497043}{396140812571321687967719751680} > 0.$$

The derivative bound follows from $|\log q| < 1$. Therefore both bracketed factors in (48) are positive and increasing.

Also $\Gamma(x+1)$ is positive and increasing for $x \geq 2$: the digamma function is increasing, and $\psi(3) = 3/2 - \gamma > 0$ using $\gamma < 1$. The logarithmic derivative of $2^{x(x-1)/2} \rho^{-x}$ exceeds $(3/2) \log 2 - \log(3/2) > 0$ for $x \geq 2$. Thus G_k and the effective envelopes are increasing on $x \geq 50$.

Finally (44) gives $C_{n,k} \leq C_{50,k}$ for all $n \leq 50$, even across an initial zero segment. Hence the explicit condition

$$y > F_{k,+}^{\text{eff}}(50) \geq C_{50,k}$$

excludes the whole earlier prefix. We obtain the global, fully specified bounds

$$\left[(F_{k,+}^{\text{eff}})^{-1}(y) \right] \leq N_k(y) \leq \left[(F_{k,-}^{\text{eff}})^{-1}(y) \right]. \quad (50)$$

These are explicit smooth equations with a concrete starting interval. Implementing numerical bounds for a chosen y would require certified interval root solving, including the needed Gamma evaluations. No implemented certified inverse solver is claimed in the present package.

12 Inverse expansions to every fixed order

Put

$$Y = \log y, \quad t = \sqrt{2Y/\ell}, \quad s = \log t, \quad c = d + 1, \quad \delta = \log(2\pi L_k/k),$$

with ℓ, β from (36). Taking the logarithm of (42) and using fixed-order Stirling gives

$$\log F_k(x) = \frac{\ell x^2}{2} + 2x \log x - \beta x + c \log x + \delta + \sum_{j \geq 1} \lambda_{k,j} x^{-j} \quad (51)$$

in the fixed-order Poincaré sense. The coefficients come from (38) and the formal logarithm of $P_k/(L_k x^d)$. For example,

$$\lambda_{k,1} = p_{k,1} + 1/6, \quad \lambda_{k,2} = p_{k,2} - p_{k,1}^2/2.$$

Theorem 12.1 (Logarithmic reversion). *For every fixed $J \geq 0$ there are recursively computable polynomials $U_0, \dots, U_J$, with $\deg U_j \leq j + 1$, such that*

$$r_k(y) = t + \sum_{j=0}^J U_j(\log t) t^{-j} + O_{k,J} \left(\frac{(\log t)^{J+2}}{t^{J+1}} \right). \quad (52)$$

The first three are

$$U_0(s) = \frac{\beta - 2s}{\ell}, \quad (53)$$

$$U_1(s) = \frac{U_0(s)^2}{2} - \frac{2U_0(s) + cs + \delta}{\ell}, \quad (54)$$

$$U_2(s) = -\frac{2U_1(s) + U_0(s)^2 + cU_0(s) + \lambda_{k,1}}{\ell}. \quad (55)$$

For general j , substitute the previous truncation plus $U_j(s)t^{-j}$ into $\log F_k(x) - \ell t^2/2$ and cancel the coefficient of t^{1-j} .

Proof. Write $x = t + u$ and expand $\log(t + u) = \log t + \log(1 + u/t)$. At order t , the residual is $\ell U_0 + 2s - \beta$, proving (53). At constant order it is

$$\ell U_1 - \ell U_0^2/2 + 2U_0 + cs + \delta,$$

which gives (54). The coefficient of t^{-1} gives (55).

In general the new unknown enters the quadratic term first as $\ell U_j t^{1-j}$. Every other contribution at that order depends only on earlier coefficients. This uniquely determines a polynomial $U_j(s)$. Inductively, products from the quadratic and $x \log x$ terms have degree at most $j + 1$ at this order; the $\log x$ and negative-power terms have no higher degree. Thus $\deg U_j \leq j + 1$.

For fixed J , the retained correction is $u = O_k(\log t)$. The finite Taylor expansions used above have uniform remainders in that range, and the fixed-order Stirling remainder is uniform on the positive real tail. After cancellation through $j = J$, the residual in the logarithmic equation is $O_{k,J}((\log t)^{J+2}t^{-J})$. Both the approximate root and the true root lie on the same final monotone interval. Using (43), or first bracketing the approximate root by a vanishing neighborhood, the mean value theorem divides this residual by a derivative comparable to ℓt . This proves (52). $\square$

In particular,

$$r_k(y) = \sqrt{\frac{2 \log y}{\log 2}} - \frac{2}{\log 2} \log \sqrt{\frac{2 \log y}{\log 2}} + \frac{\beta}{\log 2} + O_k \left(\frac{(\log \log y)^2}{\sqrt{\log y}} \right).$$

The theorem is standard asymptotic reversion applied to the target-specific counting envelope, not a new general inversion method.

If $r_{k,J}$ is the finite sum in (52), then replacing r_k in the pointwise integer bracket gives, for some $B_{k,J} > 0$ and all sufficiently large y ,

$$\left\lceil r_{k,J}(y) - B_{k,J} \frac{(\log t)^{J+2}}{t^{J+1}} \right\rceil \leq N_k(y) \leq \left\lceil r_{k,J}(y) + B_{k,J} \frac{(\log t)^{J+2}}{t^{J+1}} \right\rceil. \quad (56)$$

A finite algebraic-logarithmic truncation has the error shown here. It does not retain the exponentially small precision in (47). That precision requires solving the full Gamma-polynomial envelope equation.

Remark 12.2 ([write] The inverses against the transseries volume, 6 October 2026). The volume is `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`; its letters $\rho, \beta, \mu, \ell, t, L, X, Z, \xi$ are written here with the subscript “vol”.

- (a) *The staircase applies to $C_{n,k}$.* Let $n_1 = 1$ if $k = 1$, and for $k \geq 2$ let n_1 be the least n with $C_{n,k} > 0$ (it exists by the bouquet construction, and $n_1 \geq 2$ since $C_{0,k} = C_{1,k} = 0$). Then $(C_{n,k})_{n \geq n_1}$ is strictly increasing and unbounded: by (44), $C_{n+1,k} \geq 2^n C_{n,k} > C_{n,k}$ whenever $n \geq 1$ and $C_{n,k} > 0$, and positivity passes from n to $n + 1$. So $A_n = C_{n,k}$ satisfies the hypotheses of `p0:def:three-inverses`. For $y > C_{n_1,k}$ both $N_k(y)$ and the staircase $N_*(y)$ exceed n_1 , so they are equal, and `p0:thm:staircase(1)` gives $N_k(y) = \lceil \nu_{\mathcal{A}}(y) \rceil$ for every admissible interpolation $\mathcal{A}$ of the counts, $\nu_{\mathcal{A}} = \mathcal{A}^{-1}$.
- (b) *The brackets are analogues, not instances.* The two-ceiling brackets (46), (47), (50) and (56) are not instances of `p0:thm:staircase`: the functions $F_k, F_{k,\pm}$ and $F_{k,\pm}^{\text{eff}}$ are not admissible interpolations (they do not take the value $C_{n,k}$ at the integers, they only bound it from both sides), and each bracket is proved directly from that sandwich and monotonicity. They share the conclusion of the separation condition of `p0:thm:staircase(2)`: when the two ceilings of a bracket coincide, $N_k(y)$ is their common value; when they differ, two consecutive values remain (the two arguments of the ceilings differ by less than one there: by $O(q^r r^{-d-1})$ in (46) and (47), by $O((\log t)^{J+2} t^{-J-1})$ in (56), and in (50) because $G_k(x+1)/G_k(x) = (x+1)^{2x/\rho}$ dwarfs $(P_k + 1/10)/(P_k - 1/10)$ for $x \geq 50$; added after the independent check of 6 October 2026). The source’s remark that no single eventual rounding formula is universally valid is in line with `p0:thm:staircase(5)`. The constants B' and $B_{k,J}$ in (47) and (56) are existential; only (50) is explicit.
- (c) *Theorem 12.1 is an instance of the volume’s template only after a change of variable, and only formally.* The dominant term of $\log F_k(x)$ is quadratic, so neither `p0:thm:lambert-core` (the core $aX + b \log X$) nor `p0:prop:factorial-core` (the core $X(\kappa \log X + d)$) applies; the phase $\ell x^2/2 + 2x \log x - \beta x$ is an admissible core in the sense of `p0:def:core` (its derivative $\ell x + 2 \log x + 2 - \beta$ increases to $+\infty$), but it is neither of the two cores that Part 0 of the volume uses and inverts by Lambert W (its display `p0:eq:two-cores`). Put

$$G(x) = \sqrt{(2/\ell) \log F_k(x)},$$

defined where $\log F_k > 0$; there $\log F_k(x) = Y$ is equivalent to $G(x) = t$, since the square root is increasing. Write the expansion (51) as $(2/\ell) \log F_k(x) = x^2(1 + \varepsilon)$ with

$$\varepsilon = \frac{4 \log x}{\ell x} - \frac{2\beta}{\ell x} + \frac{2(c \log x + \delta)}{\ell x^2} + \frac{2}{\ell} \sum_{j \geq 1} \lambda_{k,j} x^{-j-2},$$

a formal series in x^{-1} of positive order with coefficients polynomial in $\log x$. Then $G(x) = x \sum_{r \geq 0} \binom{1/2}{r} \varepsilon^r = x + \frac{2}{\ell} \log x - \frac{\beta}{\ell} + B(x)$, where B collects $x\varepsilon/2$ minus its terms of order zero and the terms $r \geq 2$; each $x\varepsilon^r$ with $r \geq 2$ has order at least $r - 1 \geq 1$ in x^{-1} , so B lies in $x^{-1} \mathbb{R}[\log x][[x^{-1}]]$. Hence $G(x) = t$ is an equation of monomial–logarithmic type in the sense of `plt:def:lw-monomial-log-datum`, with $Z_{\text{vol}} = x$, $\xi_{\text{vol}} = t$ and data $(\rho_{\text{vol}}, \mu_{\text{vol}}, \beta_{\text{vol}}, B) = (1, 2/\ell, -\beta/\ell, B)$, and `plt:thm:lw-template` gives its unique formal solution $x = X_{\text{vol}} - \frac{2}{\ell} \log X_{\text{vol}} + \sum_{n \geq 1} q_n(\log X_{\text{vol}}) X_{\text{vol}}^{-n}$ with $X_{\text{vol}} = t + \beta/\ell$. Re-expanded in t and $s = \log t$ it is a formal solution of the same equation of the form $t + \sum_j U_j(s) t^{-j}$, which the triangular recursion in the proof of Theorem 12.1 determines uniquely; so the two

coincide. At the first order, $x = t + \beta/\ell - \frac{2}{\ell}s + O(s/t)$, which is (53). What the template does not give is the analytic remainder in (52), proved by the source’s mean-value argument, nor the bound $\deg U_j \leq j + 1$, which is the source’s induction (the write checked it for $j \leq 3$). The expansion of $r_k(y)$ plays the role of the volume’s asymptotic inverse (object (3) of `p0:def:three-inverses`), produced by this reversion rather than by the volume’s Lambert theorems. Equivalently, and without the square root, Theorem 12.1 is a formal instance of `p0:thm:core-reversion`, the route of Remark 11.1(a) of `a222959-zero-slope-matrices` for its own quadratic phase: with $x = t(1 + E_{\text{vol}})$, $t_{\text{vol}} = 1/t$ and $s = \log t$ an indeterminate, $(\log F_k(x) - \ell t^2/2)/t^2$ is exactly $\ell E_{\text{vol}} + \frac{\ell}{2}E_{\text{vol}}^2 + f_{\text{vol}}(t_{\text{vol}}, E_{\text{vol}})$, where

$$f_{\text{vol}}(\tau, u) = \tau[2(1+u)(s + \log(1+u)) - \beta(1+u)] + \tau^2[cs + c \log(1+u) + \delta] \\ + \sum_{j \geq 1} \lambda_{k,j} \tau^{j+2} (1+u)^{-j}$$

lies in $\tau R[[\tau, u]]$ over $R = \mathbb{R}[s]$; so $\Lambda_{\text{vol}} = \ell$, $h_{\text{vol}}(u) = \ell u^2/2$, and the triangular recurrence (`p0:eq:core-triangular`) gives $e_{j+1} = U_j$, since $x = t + \sum_{n \geq 1} e_n t^{1-n}$; for $j \leq 2$ this was checked with SymPy. (Added after the independent check of 6 October 2026.) No novelty is claimed for the inversions; the source calls Theorem 12.1 “standard asymptotic reversion”.

13 Finite accuracy and computational evidence

13.1 What the asymptotic looks like at finite sizes

The table reports the relative discrepancy $F_k(n)/C_{n,k} - 1$, using the entire dominant polynomial rather than just its leading monomial. Values are rounded diagnostics, not certified finite-error intervals.

n	$k = 1$	$k = 2$	$k = 3$	$k = 4$
5	$1.3365 \cdot 10^{-3}$	$-1.2414 \cdot 10^{-2}$	$-9.1800 \cdot 10^{-2}$	$-1.1264 \cdot 10^{-1}$
10	$3.5185 \cdot 10^{-6}$	$-2.3626 \cdot 10^{-5}$	$-1.4660 \cdot 10^{-4}$	$-7.2692 \cdot 10^{-5}$
20	$2.4401 \cdot 10^{-11}$	$-1.4342 \cdot 10^{-10}$	$-8.7540 \cdot 10^{-10}$	$6.6647 \cdot 10^{-11}$
30	$1.6922 \cdot 10^{-16}$	$-9.5481 \cdot 10^{-16}$	$-5.8314 \cdot 10^{-15}$	$1.8076 \cdot 10^{-15}$

The signs can change, as the $k = 4$ column illustrates. Nothing in the theorem promises that the dominant contribution lies on one fixed side of the true count. The safe bounds in (5) are substantially looser than these observed errors; they are designed to certify every index with elementary estimates.

13.2 Exact experiments and their limits

The standard-library checker enumerates all suppressed kernels for $k = 3, 4$, evaluates their exact rational subdivision series, verifies the rational functions (15)–(16), and exhaustively counts diagonal-one matrices through $n = 4$. It uses the multiplicative denominator and exact coefficient inversion to compute the four columns. Its displayed source comparisons through $n = 6$ are checks against OEIS, not exhaustive enumeration at $n = 6$.

An independent C++ audit uses subset zeta transforms of permutation supports to enumerate all binary matrices through $n = 5$, including $2^{25} = 33554432$ matrices at size five. Separately it enumerates all diagonal-one matrices through $n = 5$, including $2^{20} = 1048576$ at size five, and tests strong connectivity by transitive closure. It verifies the marked matching identity for every permanent value, including unattained values. At $n = 5$, selected counts are

k	$C_{5,k}$	$b_{5,k}$	$s_{5,k}$
1	3513720	29281	0
2	4339440	72324	24
3	2626800	65670	720
4	3015450	100515	4940
5	1451400	60475	13975
6	1872800	93640	26220

The independent block audit renormalizes all perfect matchings of all diagonal-one digraphs through $n = 4$ and verifies that all 16774 matching normalizations preserve their SCC-size/permanent signature. It also enumerates complete signatures through $n = 5$, $k \leq 4$, and an independent exact marked transform reproduces them. For instance, at $n = 5$, $k = 4$, the signatures and diagonal-one counts are

Multiset of (permanent, size)	Count
$\{(2, 2), (2, 2)\}$	18735
$\{(2, 2), (2, 3)\}$	2540
$\{(4, 3)\}$	34620
$\{(4, 4)\}$	39680
$\{(4, 5)\}$	4940

Their sum is 100515, as required. These finite tests support the implementation; the all-size proofs are the combinatorial and analytic arguments above.

13.3 Reproducibility contract

The ZIP contains the manuscript, code, reference outputs, and build instructions. Mandatory Python computation uses only the standard library, exact integers and fractions for all certificates, and explicit exceptions rather than optimization-removable assertions for critical checks. The normal and optimized Python runs must reproduce the reference certificate bytes. Malformed certificate-domain inputs are tested in both modes.

The optional C++ exhaustive programs and their reference outputs are identified separately from mandatory Python replay. Numerical diagnostics have no role in the proofs or the certified intervals. The build requires a local TeX installation and compiles without shell escape. The package records its file inventory and hashes and is intended to reproduce the same PDF bytes on the same installed Python/TeX stack; byte identity across different TeX versions is not promised. Detailed commands, guard tests, and the exact mandatory/optional distinction are in the packaged README files. No network access, downloaded third-party source code, or third-party paper files are needed for replay.

[write] Reruns at intake, 6 October 2026. On copies of the delivered layout (Windows 11, Python 3.14.4), at placement and again at the write: `reproduce.py` stops with “normal byte mismatch: exact_checks.json”, because the delivered scripts write their JSON in text mode and Windows emits CRLF line endings (5,356 against 5,158 bytes; equal after conversion). Run through a wrapper that only converts CRLF to LF before the byte comparison, it passes: all five reference files are reproduced byte for byte in normal and optimized Python, and the guards pass (18 negative and 3 positive tests), in about 20 s. `test_build.py` passes 171 tests on Windows; the delivered receipt `data/generated-build_guards.json` records 176, and the five missing are FIFO and POSIX-only rejection cases that the suite skips there. At placement only, the optional C++ audits were compiled with g++ 16.1.0 (WinLibs); they match the delivered TSV files (after removing carriage returns) only when linked with `-static`; dynamically linked builds crash on this machine, so `reproduce.py`

-optional-cpp does not run as delivered here. The 16,774 matching normalizations pass, and the 2^{25} matrices at $n = 5$ take about 1.2 s. The PDF and ZIP builder was not run. The report README gives the rerun routes.

[write] **Independent check of the write, 6 October 2026.** An adversarial check made by the intake after the write (commit 71813ce3f) re-derived the statements marked [write] with its own code, and fetched the OEIS entries again (the revisions quoted in Remark 1.4 are current; Royle’s row $n = 6$ is as quoted). *Remark 1.4.* ρ and a at 130 digits (a root of the 110-term series; a also as $\rho E(\rho/2)$, difference 10^{-131}) agree with all 103 printed digits of A245654 and A245655. With A003024 from its own recurrence, $A003024(n) a \rho^n / (n! 2^{\binom{n}{2}}) - 1$ is $-1.3 \cdot 10^{-3}$, $-3.5 \cdot 10^{-6}$, $-9.3 \cdot 10^{-9}$, $-2.4 \cdot 10^{-11}$, $-6.4 \cdot 10^{-14}$ at $n = 5, 10, 15, 20, 25$. Every matrix of permanent one with $n \leq 4$ (1, 6, 150, 13032 of them) has pairwise distinct rows and columns; the row and the column orbits number 1, 3, 25, $543 = A003024(n)$, the joint orbits 1, 2, 6, 31. *The table of Section 1.* A C++ program counted all 2^{n^2} binary $n \times n$ matrices for $n \leq 6$ (all 2^{36} at $n = 6$, as weighted row multisets, the totals checked against 2^{n^2}) by an exact permanent, without the transform: rows $n \leq 6$, $k \leq 4$ equal the table, the OEIS data and Royle’s row, including $T(6, 0) = 13906734081$. A second program counted the strongly connected kernels directly for $m \leq 7$: $s_{m,2} = (m-1)!$, $s_{m,3} = 0, 9, 84, 720, 6480, 63000$ and $s_{m,4} = 0, 6, 256, 4940, 80400, 1259160$ for $m = 2, \dots, 7$, which are $m![z^m]$ of (15) and (16). *Section 7.* The thirteen certified quantities by a third route (exact Taylor coefficients of $e^{-z}W_k$ from the printed rational functions, 110 terms at 130 digits, Laurent division at ρ): each lies inside its printed interval, with margins at least $1.7 \cdot 10^{-27}$; $c_{2,2} = h_2/a^2$ and $c_{4,3}/2 = h_2^2/(2a^3)$ to 30 digits. *Remark 12.2.* (a) and (b) re-derived, the equation numbers checked in the PDF; SymPy reproduces U_0, U_1, U_2 and $\deg U_3 = 4$, and the expansion of G in (c). No mathematical error was found. Three additions, each marked with the date: Remark 1.4(d) names the number of joint orbits (A003087) with its proof; Remark 12.2(b) says why at most two consecutive values remain; and (c) records that Theorem 12.1 is also a direct formal instance of **p0:thm:core-reversion**, as in the sibling report **a222959-zero-slope-matrices**. The digit comparison of Remark 1.4(c) is extended to all 103 digits.

14 Source review and attribution boundary

14.1 The live sequence and the intended model

The A089479 internal record was read on 3 October 2026, including its revision 31 of 19 February 2026, definitions, initial rows, comments and references. It counts all labelled binary square matrices by their ordinary integer permanent. The displayed record did not state the fixed-positive- k asymptotic proved here. A089482 supplies the established permanent-one/DAG correspondence, and A003024 supplies the classical DAG sequence. This current-record observation is not a conclusion about all literature.

Different nearby models are intentionally excluded. Matrices with exactly two ones per row, zero-permanent matrices, degree-constrained permanent expectations, and counts of distinct permanent values attained by a matrix family are not the same as the fixed-positive-permanent fibers $C_{n,k}$.

14.2 Primary results inspected and their relation

1. De Panafieu and Dovgal, *Symbolic method and directed graph enumeration* [4]. The seven-page text, including Theorem 3.4 and its proof, was read. Its exact SCC-family transform, normalization, and marked version are classical input here. The authors explicitly credit Robinson’s earlier machinery. The use of this transform is not a novelty claim.
2. De Panafieu and Dovgal, *Counting directed acyclic and elementary digraphs* [5]. The model, exact elementary formula in Section 2.3, and the complete relevant asymptotic theorem statements were inspected. Their elementary class has only singleton and directed-cycle SCCs,

including 2-cycles. Its main asymptotic regime constrains the edge count near $n(1 + \mu n^{-1/3})$. The $k = 2$ family here is a one-cycle specialization of that classical exact class. Its mere generating-function construction is therefore not presented as new.

3. Dovgal and Nurligareev, *Asymptotics for graphically divergent series: dense digraphs and 2-SAT formulae* [6]. The model, transfer scale, SCC transform, and theorem/model statements of Sections 4.3–4.4 in the August 2026 version were inspected. The printed fixed-total-SCC-count results have one large SCC and a fixed number of singleton components. Fixed positive permanent here instead gives $n - O(1)$ SCCs in the limiting size law. Their framework is relevant prior; the inspected statements do not supply the $\Omega(k)$ degree and amplitude formula of this report.
4. Kim, Lee and Seol, *Matrices with prescribed permanent using binary number system of integers* [7]. The full six-page paper was read. It constructs matrices attaining prescribed permanent values and analyzes a particular family; its final section concerns the number of distinct attained values. It does not give an asymptotic for the number of all binary matrices having one fixed permanent. Its existence results remain related classical work.
5. Greenhill, Hasheminezhad, Iliffe and McKay, *Asymptotic enumeration of constrained bipartite, directed and oriented graphs by degree sequence* [8]. The model and relevant Section 4, including Theorems 4.2–4.5 and proofs, were read. These concern expected permanents under prescribed degree conditions and near-complete pointwise estimates. An expectation under degree conditioning is different from the fixed-permanent fiber count considered here.

The full text of Robinson’s *Counting digraphs with restrictions on the strong components*, linked from A350790, could not be retrieved from its author’s server during the source pass. The needed general transform was instead read in full in the primary 2019 restatement. This is a historical-priority inspection limitation; no claim is made that the inaccessible text lacks a further fixed-permanent theorem.

Bounded exact-ID and thematic searches of the available report collections found no matching report. Potentially nearby reports on bounded-indegree DAGs, lonesum-decomposable binary matrices, orientation partition sums, and cyclic-word cover models were checked at their actual model level and concern different classes. These checks do not exhaust unpublished work or the literature worldwide.

[write] **Added 6 October 2026.** The write reread items 1–3 and 5 at the places named (details in Section 1.2) and found them described correctly; it did not read item 4. Robinson’s paper was again unreachable. The reports named in the last paragraph are `a397711-bounded-indegree-dags`, `a299907-lonesum-decomposable-matrices`, `a372395-acyclic-orientation-partitions` and `a108242-regular-cyclic-word-covers`; the case $k = 1$ is in addition an instance of the asymptotic stated in OEIS A003024, which the source cites for the correspondence only (Remark 1.4).

14.3 What is proved here and what is standard

Classical ingredients include the matching/permanent correspondence, circulation decomposition, SCC condensation, labelled subdivision counting, Robinson’s graphic component transform, Rouché and Cauchy arguments, finite-pole transfer, Stirling expansion, and inverse reversion. This report supplies the fixed-permanent finite-kernel reduction, the explicit multiplicative marking formulas, the transformed-prime positivity identity, the exact prime-factor pole degree and leading amplitude, low- k prefactors and certificates, and the stated canonical block and threshold consequences. This

attribution separates the application-specific derivation from its established tools without asserting worldwide priority.

15 Further questions

The results leave several concrete directions open.

1. **Efficient large-permanent kernel algorithms.** The finite bounds prove computability, but direct multigraph enumeration grows quickly. A canonical-kernel or dynamic program tailored to $\text{per}(I + M) = k$ could make additional columns practical.
2. **Sharper small-permanent error constants.** The circle $|z| = 2$ and the majorants used here are intentionally simple. Certified isolation of the next zeros, or optimized circle estimates, could reduce both exponential and constant factors in the low- k remainder.
3. **Explicit correction laws.** The weighted-measure proof gives all corrections algorithmically. Computing the first signed measure for a composite k , especially $k = 4$ or 6 , would quantify the finite-size probability of a composite-permanent block and the interaction of repeated prime types.
4. **Certified inverse implementation.** The effective equations already give rigorous targets. A practical interval solver for the Gamma-polynomial envelopes, together with exact prefix counts, would turn them into certified numerical threshold answers for large specified y .
5. **Growing permanent regimes.** None of the fixed- k estimates controls $k = k(n)$. Determining where the finite-kernel picture loses uniformity is a separate problem, requiring bounds on both kernel counts and prime amplitudes.
6. **A fuller historical comparison.** Obtaining the inaccessible Robinson component paper and examining further matching-theoretic enumeration literature would strengthen the bounded prior review. This is an attribution question distinct from the proved correctness of the present formulas.

[write] **Added 6 October 2026, batch 108.** Following Vladimir Reshetnikov’s standing rule of 4 October 2026, every claim of the source that is not proved in full is stated as an open question with its source, the argument offered and what is missing. The six questions above are the source’s and stand as written; its non-claims are collected in Section 1.2. The intake found no false mathematical claim in the source, and the write found none either: every proof was rechecked, and the computations behind (15)–(16) and the table of Section 7 were redone independently (Section 1.2). One claim is not supported by the package and is added below. Dated remarks on the source’s questions: on Question 2, the write’s floating values of the $c_{k,j}$ show how loose the integer bounds of Section 8 are (for instance $|c_{2,1}| = 4.82$ against the bound 5, but $|c_{4,1}| = 2.37$ against 3), while most of each M_k comes from the circle majorants $|D_k|$, not from the principal parts (the principal parts contribute 33, 60, 99); on Question 4, Remark 12.2(b) records that only the brackets (50) are explicit; on Question 6, Robinson’s paper was again unreachable on 6 October 2026 (Section 14).

7. **The second implementation** (source, Section 7.3; added by the write). The source states that “a second independent rational implementation using 50 terms and formal Laurent-series inversion reproduces the displayed intervals”. No such implementation is in the package, so the statement cannot be checked from it. Nothing in the report depends on it: the intervals are certified by the shipped primary program, whose output the intake reproduced, and the write’s floating computation (note at the end of Section 7.3) agrees with all thirteen intervals. What is missing is the independent rational certificate itself: a second exact enclosure of ρ , a , h_2 ,

h_3 , $H_4(\rho)$ and the $c_{k,j}$, by Laurent inversion rather than by the derivative formulas (29)–(30), written and published with its code.

References

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- [7] I.-P. Kim, S.-G. Lee and H.-G. Seol, *Matrices with prescribed permanent using binary number system of integers*, International Journal of Pure and Applied Mathematics **19**(3) (2005), 413–418. <https://www.ijpam.eu/contents/2005-19-3/12/12.pdf>.
- [8] C. Greenhill, M. Hasheminezhad, I. Iliffe and B. D. McKay, *Asymptotic enumeration of constrained bipartite, directed and oriented graphs by degree sequence*, arXiv:2601.04822v1, 8 January 2026. <https://arxiv.org/abs/2601.04822>.
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- [10] The OEIS Foundation Inc., *A245654*, decimal expansion of the smallest positive root of $\sum_{n \geq 0} (-1)^n x^n / (2^{n(n-1)/2} n!)$, revision #19, read 6 October 2026 (added by the write). <https://oeis.org/A245654>.
- [11] The OEIS Foundation Inc., *A245655*, decimal expansion of a constant in the asymptotics of labeled acyclic digraphs, revision #15, read 6 October 2026 (added by the write). <https://oeis.org/A245655>.
- [12] The OEIS Foundation Inc., *A003087*, number of acyclic digraphs with n unlabeled nodes, revision #71, read 6 October 2026 (added after the independent check). <https://oeis.org/A003087>.